

3(a)(i)	$mg = kx$ $(0.5)(9.81) = k(0.148 - 0.13)$ $k = 272.5 \approx 273 \text{ N m}^{-1}$	[1] ans
(ii)	Above C, the weight of mass provides the centripetal force. $mg = mr\omega^2$ $9.81 = 0.13\omega^2$ $\omega = \sqrt{\frac{9.81}{0.13}} = 8.687 \approx 8.7 \text{ rad s}^{-1}$	[1] state [1] sub
(iii)	$T - mg = mr\omega^2$ $(272.5)(x) - (0.50)9.81 = (0.50)(0.13 + x)(8.7)^2$ $234.655x = 9.82485$ $x = 0.0418 \text{ m}$ $L = 0.13 + 0.0418 = 0.172 \text{ m}$	[1] sub [1] ans
(b)(i)1.	$F_c = mr\omega^2$ $= (5)(6.4 \times 10^6) \left(\frac{2\pi}{(24)(3600)} \right)^2$ $= 0.169 \text{ N}$	[1] sub [1] ans

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Suggested Solutions with Markers' Comments

Qn	Suggested solution	Remarks
2.	$N = W - F_c$ $N = (5)(9.81) - 0.169$ $= 48.9 \text{ N}$	[1] sub [1] ans
(ii)	At the poles, the mass is not moving in a circle. Hence normal contact force equals gravitational force. ($N = F_G$) <i>[At the equator, some of the gravitational force is used to provide the centripetal force to move the mass in a circle. ($N = F_G - F_c$)]</i>	[1] [1]
4(a)	On the surface of the Earth	